

Muhammad Umair Hassan

SENIOR DATA SCIENTIST

Hagaløkkveien 26, 1383, Asker, Norway

+47 4866 3587 | mumairhsn@gmail.com | mumairhassan.github.io | [mumairhsn](https://www.linkedin.com/in/mumairhsn) | [@mumairhsn](https://twitter.com/mumairhsn)

Objective

Highly motivated and skilled computer science professional with a strong background in artificial intelligence, computer vision, data science, internet-of-things, and machine learning. With expertise in digital tools and technologies, I aim to develop innovative, sustainable solutions and bridge the gap between IT and engineering. Adaptable and creative problem-solver with a strong background in engineering who can successfully deliver projects and coherently communicate project findings.

Education

Norwegian University of Science and Technology-(NTNU)

Norway

PHD COMPUTER SCIENCE

2020 - 2024

- **Dissertation Title:** Advancing Digital Twin Creation for Smart Cities: Innovative Approaches in Visual Representation and Simulation

University of Jinan

P.R. China

MS COMPUTER SCIENCE AND ENGINEERING

2017 - 2020

- **Dissertation Title:** Instance Retrieval Based on Combination of Geometric Features and CNN

University of the Punjab

Pakistan

BS COMPUTER SCIENCE

2013 - 2017

Professional Experience

- Aug, 2026 - Present **Senior Data Scientist**, 4Subsea, Norway (**Permanent Employment**)
- Apr, 2025 - Aug, 2026 **Researcher in Machine Learning**, Norwegian University of Life Sciences (NMBU), Norway (**Project-based Employment**)
- May, 2024 - Aug, 2026 **Research Scientist in Computer Vision and Artificial Intelligence**, Norsvin SA, Norway (**Permanent Employment**)
- Nov, 2023-Apr, 2024 **Assistant Professor**, Taught Web Technologies course at the Department of ICT and Natural Sciences, Norwegian University of Science and Technology, Norway (**Contractual Employment**)
- 2018-2020 **Research Assistant**, Shandong Provincial Key Laboratory of Network-Based Intelligent Computing, University of Jinan, P.R. China

Projects

Heart Segmentation of Non-Contrast Pigs Data

Norsvin SA, Norway

PYTHON & DEEP LEARNING

- Developed a CT heart image segmentation pipeline for CT scanned pigs data using Transfer Learning based nnU-Netv2.
- Implemented automated 5-fold cross-validation and GPU-accelerated training with torch.
- Standardizing CT images of pigs scanned from two different scanners.
- **Technologies Used:** Python, PyTorch, nnU-Net, CUDA, 3D CNN, Transfer Learning

CT Image Segmentation for Osteochondrosis Data

Norsvin SA & NMBU, Norway

PYTHON & DEEP LEARNING

- Developed a CT image segmentation production pipeline for CT scanned pigs data using nnU-Net and customized 3D transformer architectures by achieving state-of-the-art F1 score compared to existing models.
- Implemented automated 5-fold cross-validation and GPU-accelerated training with torch.
- Engineered advanced preprocessing and data augmentation strategies to boost segmentation performance.
- **Technologies Used:** Python, PyTorch, nnU-Net, CUDA, 3D CNN, 3D Vision Transformer

Pig Detection, Tracking and Behaviour Analysis

Norsvin SA, Norway

PYTHON & DEEP LEARNING

- Implementation of pig posture detection in pig barns by improving the accuracy upto 98%
- Handling pig tracking using sorting algorithms
- Preparing and handling datasets for the detection and behaviour recognition analysis
- **Technologies Used:** YOLOv8, SlowFast, Google Cloud Platform, PyTorch, Containerization

Automated Traffic Signal Control Using Real-Time CCTV Analysis

NTNU, Norway

PYTHON & DEEP LEARNING

- Developed an innovative algorithm for dynamic traffic signal control, leveraging real-time data analysis
- Utilized Faster R-CNN for accurate vehicle detection in diverse traffic conditions
- Created a specialized local dataset from these live feeds for algorithm training and validation
- **Technologies Used:** Faster R-CNN, CCTV Analytics, Real-Time Data Processing

Face Anonymization

NTNU, Norway

PYTHON & DEEP LEARNING

- Developed an algorithm for deep privacy preservation of human subjects using Jetson Nano
- Fine-tuning of YOLOv7 & YOLOv5 to work on edge devices for real-time anonymization of data

Tyre Defect Detection using GLCM, Gabor Filter and Radiography

University of Jinan, China

MATLAB & C++

- Developed a method for identifying defects in tyre textures using Gray-Level Co-occurrence Matrix (GLCM) and Gabor filter texture segmentation
- Enhanced defect detection accuracy by matching abnormal textures with standard images and using GLCM features for scoring
- Implemented advanced image processing techniques including radiographic analysis, curvelet transformation, and canny edge detection to decompose and examine tyre defects
- **Technologies Used:** MATLAB, C++, Gabor Filter Texture Segmentation

Asymmetric Hashing based on Generative Adversarial Network – AGAN

University of Jinan, China

PYTHON & DEEP LEARNING

- Proposed AGAN framework that provides binary representations with an accurate image retrieval ability
- Developed an advanced sign-activation and loss function-based learning process by designing a contemporary model based on an encoder loss, a generator loss, and a discriminator loss
- Hashing-based strategy resulted in efficient training time for large-scale nearest neighbour searching

CNN-based Fusion of Salient Objects

University of Jinan, China

MATLAB, PYTHON & DEEP LEARNING

- Proposed a convolutional neural network based strategy to fuse salient objects information and generating a high-level model
- The development was done using MATLAB for salient object detection and segmentation whereas Python was used for deep learning assignment to combine dual information

Shape Matching of Non-rigid Structures

University of Jinan, China

MATLAB

- Proposed an automatic dense correspondence method to match the mesh vertices of two 3D shapes under near-isometric and non-rigid deformations
- The proposed method is implemented MATLAB and combines three different types of graphical information

Book Corner App

University of the Punjab,
Pakistan

ANDROID & JAVA

- Led the development of a high-performance Android application using Java, focusing on optimizing user experience and interface design for mobile devices.
- Engineered key features, including push notifications, real-time data synchronization, and offline access capabilities for book-shop retailers, significantly enhancing app functionality and user engagement.
- Implemented best practices in Java for memory management and efficient data processing, resulting in a smooth and responsive application.

Online Shopping & Cart System

University of the Punjab,
Pakistan

ASP .NET, C# & SQL

- Led designing and implementing a user-friendly online shopping platform, ensuring a responsive design for seamless access across devices.
- Engineered an SQL database schema for efficient storage and retrieval of product, user, and transaction data.
- Integrated secure payment processing with support for multiple payment methods to enhance transaction safety and reliability.
- Implemented a comprehensive user authentication system, including secure login, registration, and password recovery functionalities.

Student Web Portal System

University of the Punjab,
Pakistan

ASP .NET, C# & SQL

- Led the creation of a comprehensive web portal to facilitate student access to academic resources, course registrations, and personal academic records.
- Implemented essential features such as course enrollment, grade tracking, and academic calendar management, enhancing the educational experience for students.
- Designed an intuitive and accessible user interface, ensuring easy navigation and a positive user experience for students and faculty.

Skills

Programming: Python, R, C/C++, C#, HTML, CSS, MATLAB, Java, Javascript, SQL

Public Library: PyTorch, Tensorflow, Scikit-image, CUDA, Keras, Scipy, Numpy, pandas, Caffe, OpenCV, nibabel, SimpleITK

Environments & IDE: Linux, Github, BitBucket, Docker, Bash shell, Windows, Visual Studio, NetBeans, Google Cloud Platform, PyCharm, Azure, Microsoft Intune, Microsoft 365 tools

Languages: English (fluent), Norwegian (basic), Mandarin (basic), Urdu (native)

JD-related Skills:

Experienced in working with large datasets using TensorFlow and PyTorch.

In-depth understanding of various AI techniques and their applications.

Deploying machine learning models on Linux servers and developing in Linux-based environments.

Experience with research and development methods.

Excellent analytical and problem-solving abilities.

Publications

PUBLISHED (JOURNAL)

Nordbø, Ø., **Hassan, M. U.**, Gangsei, L. E., Grindflek, E. & Olstad, K. (2025). Automated Identification of Bones and Joints in Whole-body CT-scan Images of Pigs. Journal of Animal Science. (IF: 2.9)

Hassan, M. U., Alaliyat, S., & Hameed, I. A. (2024). Toward the Creation of a Digital Twin Authoring Tool: A Smart Mobility Perspective in Smart Cities. IEEE Access. (IF: 3.6)

- Hassan, M. U.**, Zhao, X., Sarwar, R., Aljohani, N. R., & Hameed, I.A. (2024). SODRet: Instance retrieval using salient object detection for self-service shopping. *Machine Learning with Applications*. (IF: 4.9)
- Sarwar, R., Perera, M., Teh, P. S., Nawaz, R., & **Hassan, M. U.** (2024). Crossing linguistic barriers: authorship attribution in Sinhala texts. *ACM Transactions on Asian and Low-Resource Language Information Processing*. (IF: 2.0)
- Sarwar, R., Ha, L. A., ... & **Hassan, M. U.** (2024). AGI-P: A Gender Identification Framework for Authorship Analysis Using Customized Fine-Tuning of Multilingual Language Model. *IEEE Access*. (IF: 3.6)
- Abbas, S. K., Khan, M. U. G., **Hassan, M. U.**, & Hameed, I. A. Vision Based Intelligent Traffic Light Management System using Faster R-CNN. *CAAI Transactions on Intelligence Technology*. (IF: 7.3)
- Hassan, M. U.**, Alaliyat, S., & Hameed, I. A. (2023). Image Generation Models from Scene Graphs and Layouts: A Comparative Analysis. *Journal of King Saud University - Computer and Information Sciences*. (IF: 6.1)
- Hassan, M. U.**, Alaliyat, S., Sarwar, R., Nawaz, R., & Hameed, I. A. (2023). Leveraging Deep Learning and Big Data to Enhance Computing Curriculum for Industry-Relevant Skills: A Norwegian Case Study. *Heliyon*. (IF: 3.6)
- Hassan, M. U.**, Steinnes, O. H., Gustafsson, E., Løken, S., & Hameed, I. A. (2023). Predictive Maintenance of Norwegian Roads Network Using Deep Learning Models. *Sensors*. (IF: 3.5)
- Zaman, F., Afzal, M., Teh, P. S., Sarwar, R., Kamiran, F., Aljohani, N. R., Nawaz, R., **Hassan, M. U.**, & Sabah, F. (2024) Intelligent Abstractive Summarization of Scholarly Publications with Transfer Learning. *Journal of Informatics and Web Engineering* 3, no. 3 (2024): 256-270.
- Hussain, S. A., **Hassan, M. U.**, Nasar, W., Ghorashi, S., ... & Hameed, I. A. (2023). Efficient Trajectory Clustering with Road Network Constraints Based on Spatiotemporal Buffering. *ISPRS International Journal of Geo-Information*. (IF: 2.8)
- Bhakar, S., Sinwar, D., Pradhan, N., ... & **Hassan, M. U.** (2023). Computational Intelligence-Based Disease Severity Identification: A Review of Multidisciplinary Domains. *Diagnostics*. (IF: 3.3)
- Zhang, M., **Hassan, M. U.**, Niu, D., Zhao, X., Hameed, I. A. & Hassan, S. U. (2022). A methodology for shape matching of non-rigid structures based on integrated graphical information. *Displays*. (IF: 3.4)
- Yaqoob, I., **Hassan, M. U.**, Niu, D., Zhao, X., Hameed, I. A., & Hassan, S. U. (2022). A novel person re-identification network to address low-resolution problem in smart city context. *ICT Express*. (IF: 4.2)
- Hassan, M. U.**, Niu, D., Zhang, M., & Zhao, X. (2022). Asymmetric hashing based on generative adversarial network. *Multi-media Tools and Applications*, 1-17. (IF: 3.00)
- Yan, A., Chen, Z., Zhang, H., Peng, L., Yan, Q., **Hassan, M. U.**, ... & Yang, B. (2021). Effective detection of mobile malware behavior based on explainable deep neural network. *Neurocomputing*, 453, 482-492. (IF: 6.5)
- Hassan, M. U.**, Yaqoob, I., Zulfiqar, S., & Hameed, I. A. (2021). A comprehensive study of HBase storage architecture—A systematic literature review. *Symmetry*, 13(1), 109. (IF: 2.2)
- Gribbestad, M., **Hassan, M. U.**, Hameed, I. A., & Sundli, K. (2021). Health monitoring of air compressors using reconstruction-based deep learning for anomaly detection with increased transparency. *Entropy*, 23(1), 83. (IF: 2.0)
- Gribbestad, M., **Hassan, M. U.**, & Hameed, I. A. (2021). Transfer learning for Prognostics and health Management (PHM) of marine Air Compressors. *Journal of Marine Science and Engineering*, 9(1), 47. (IF: 2.8)
- Li, M., Zhang, M., Niu, D., **Hassan, M. U.**, Zhao, X., & Li, N. (2020). Point set registration based on feature point constraints. *The Visual Computer*, 36(9), 1725-1738. (IF: 2.9)
- Hassan, M. U.**, Karim, S., Shah, S. K., Abbas, S., Yasin, M., Shahzaib, M., & Umair, M. (2018). A Comparative Study on Frequent Link Disconnection Problems in VANETs. *EAI Endorsed Transactions on Energy Web*, 5(17), e2-e2.
- Shaukat, K., **Hassan, M. U.**, Masood, N., & Shafat, A. B. (2017). Stop words elimination in Urdu language using finite state automaton. *International Journal of Asian Language Processing*, 27(1), 21-32.

PUBLISHED (CONFERENCE)

- Hassan, M. U.**, Nordbø, Ø. Olstad, K., Aasmundstad, T. (2026) Automated osteochondrosis detection from CT scanned pigs; genetic parameters and comparison with manual scoring, 2026 World Congress on Genetics Applied to Livestock Production, Wisconsin, USA. (Accepted)
- Hassan, M. U.**, Olstad, K., Sevellano, C. A., Grindflek, E., Nordbø, Ø. (2026) Automated Detection of Osteochondrosis in Live Pigs Using the nnU-Netv2 Deep Learning Model, 58th Swine days' Research, Saint-Malo, France.

- Hassan, M. U.**, Stava, M., & Hameed, I. A. (2023). Deep Privacy based Face Anonymization for Smart Cities, In 2023 International Conference on Smart Applications, Communications and Networking (SmartNets), Istanbul, Turkiye, 2023, pp. 1-6.
- Hassan, M. U.**, Angelaki, S., Alfaro, C. V. L., ... & da Silva Torres, R. (2022, June). Digital Twins for Lighting Analysis: Literature Review, Challenges, and Research Opportunities. In 36th International ECMS Conference on Modelling and Simulation, ECMS 2022 (Vol. 36, No. 1, pp. 226-235).
- Hassan, M. U.**, Zafar, N., Ali, H., Yaqoob, I., Alaliyat, S. A. A., & Hameed, I. A. (2022). Collaborative Filtering Based Hybrid Music Recommendation System. In Proceedings of International Conference on Information Technology and Applications (pp. 239-249). Springer, Singapore.
- Cui, L., Zhao, W., **Hassan, M. U.**, & Yaqoob, I. (2020, December). Shape Matching Based on the Enhancement of Riemannian Structure Information. In Proceedings of the 2020 4th International Conference on Vision, Image and Signal Processing (pp. 1-5).
- Yaqoob, I., **Hassan, M. U.**, Niu, D., Irfan, M. M., Zafar, N., & Zhao, X. (2020, December). Efficient Deep Learning Approach to Address Low-Resolution Person Re-Identification. In Proceedings of the 2020 4th International Conference on Vision, Image and Signal Processing (pp. 1-5).
- Ma, Y., Zhang, J., Niu, D., **Hassan, M. U.**, & Zhao, X. (2019, December). An Unsupervised Approach for 3D Medical Image Registration. In Proceedings of the 2019 7th International Conference on Information Technology: IoT and Smart City (pp. 259-263).
- Hassan, M. U.**, Niu, D., Zhao, X., Shohag, M. S. A., Ma, Y., & Zhang, M. (2019, December). Salient object detection based on CNN fusion of two types of saliency models. In 2019 International Conference on Image and Vision Computing New Zealand (IVCNZ) (pp. 1-6). IEEE.
- Shabir, M. A., **Hassan, M. U.**, Yu, X., & Li, J. (2019, November). Tyre defect detection based on GLCM and gabor filter. In 2019 22nd International Multitopic Conference (INMIC) (pp. 1-6). IEEE.
- Shabir, M. A., **Hassan, M. U.**, Yu, X., & Li, J. (2019, November). Extensive Techniques to Detect Defects in Tyres through Radiography. In 2019 22nd International Multitopic Conference (INMIC) (pp. 1-4). IEEE.
- Hassan, M. U.**, Shohag, M. S. A., Niu, D., Shaukat, K., Zhang, M., Zhao, W., & Zhao, X. (2019, August). A framework for the revision of large-scale image retrieval benchmarks. In Eleventh International Conference on Digital Image Processing (ICDIP 2019) (Vol. 11179, pp. 1154-1161). SPIE.
- Zhang, M., **Hassan, M. U.**, Niu, D., Li, N., Liu, M., Zhou, J., & Zhao, X. (2019, August). Shape correspondence based effective combination of linear and quadratic assignment matrices. In Eleventh International Conference on Digital Image Processing (ICDIP 2019) (Vol. 11179, pp. 1162-1170). SPIE.
- Shohag, M. S. A., **Hassan, M. U.**, Niu, D., Kong, X., Zhao, X., & Rahman, F. (2019, May). Graph Based Image Matching Using the Fusion of Several Kinds of Features. In Proceedings of the 2019 4th International Conference on Multimedia Systems and Signal Processing (pp. 188-193).
- Hassan, M. U.**, Shahzaib, M., Shaukat, K., Hussain, S. N., Mubashir, M., Karim, S., & Shabir, M. A. (2019). DEAR-2: An energy-aware routing protocol with guaranteed delivery in wireless ad-hoc networks. In Recent Trends and Advances in Wireless and IoT-enabled Networks (pp. 215-224). Springer, Cham.
- Yan, A., Chen, Z., Wang, L., Peng, L., **Hassan, M. U.**, & Zhao, C. (2018, December). Neural Network Rule Extraction for Real Time Traffic Behavior Identification. In 2018 International Conference on Security, Pattern Analysis, and Cybernetics (SPAC) (pp. 146-151). IEEE.
- Shen, J., Chen, Z., Wang, S., Zhu, Y., & **Hassan, M. U.** (2018, July). DroidDetector: a traffic-based platform to detect android malware using machine learning. In Third International Workshop on Pattern Recognition (Vol. 10828, pp. 160-168). SPIE.
- Ma, Y., **Hassan, M. U.**, Niu, D., & Wang, L. (2018, July). Glandular cavity segmentation based on local correntropy-based K-means (LCK) clustering and morphological operations. In Third International Workshop on Pattern Recognition (Vol. 10828, pp. 108-114). SPIE.
- Hassan, M. U.**, Shaukat, K., Niu, D., Mahreen, S., Ma, Y., Haider, F., ... & Zhao, X. (2018, May). An Overview of Schema Extraction and Matching Techniques. In 2018 2nd IEEE Advanced Information Management, Communicates, Electronic and Automation Control Conference (IMCEC) (pp. 1290-1294). IEEE.
- Hassan, M. U.**, Shaukat, K., Niu, D., Mahreen, S., Ma, Y., Zhao, X., & Shabir, M. A. (2018, May). Web-Logs Prediction with Web Mining. In 2018 2nd IEEE Advanced Information Management, Communicates, Electronic and Automation Control Conference (IMCEC) (pp. 1295-1299). IEEE.

Ma, Y., **Hassan, M. U.**, Niu, D., & Wang, L. (2017, November). The segmentation of glandular cavity based on K-means and mathematical morphology. In 2017 4th International Conference on Systems and Informatics (ICSAI) (pp. 1287-1291). IEEE.

Dar, K. S., Shafat, A. B., & **Hassan, M. U.** (2017, June). An efficient stop word elimination algorithm for Urdu language. In 2017 14th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON) (pp. 911-914). IEEE.

Awards, Fellowships, & Grants

2017-2020	Chinese Government Scholarship For Master Studies , Ministry of Education, P.R. China	<i>Fully Funded</i>
2019	Machine Learning Research School , VISTEC, Bangkok, Thailand	<i>Fully Funded</i>
2018	Student Best Paper Award , IEEE IMCEC, Xian, P.R. China	
2017	3rd Position in SISE Graduate Academic Research Competition , University of Jinan, P.R. China	<i>RMB 3000</i>

Presentations

Delivered Oral Presentation at 1st Artificial Intelligence for Animal Science conference 2025, Zürich, Switzerland

Delivered Oral Presentation at IEEE SmartNets 2023, Istanbul, Türkiye

Delivered Oral Presentation at 36th ECMS Conference on Modelling and Simulation, Norway

Delivered Oral Presentation at 34th Image and Vision Computing, New Zealand

Poster Presentation at Machine Learning Research School, Bangkok, Thailand

Delivered Oral Presentation at 22nd INMIC 2019, Islamabad, Pakistan

Delivered Oral Presentation at 11th ICDIP 2019, Guangzhou, China

Delivered Oral Presentation at IEEE IMCEC 2018, Xian, China

Delivered Oral Presentation at EAI FUTURE5V 2017, Islamabad, Pakistan

Online Courses

- 2019 **IBM Cognitive Class**, Python for Data Science
- 2019 **IBM Cognitive Class**, Machine Learning with Python
- 2019 **IBM Cognitive Class**, Deep Learning Fundamentals
- 2019 **IBM Cognitive Class**, Deep Learning with TensorFlow

INTERESTS

Keeping up-to-date with the latest trends and advancements in AI, computer vision and data science.

Playing badminton and cooking different cuisines.

REFERENCES

Available upon request.